

Sungjin Nam

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[Google Scholar](#) · [LinkedIn](#)

Applied Research Scientist

Experienced applied research scientist leveraging natural language processing (NLP) and machine learning (ML) techniques to build automated operational solutions and human-centered applications. Fluently delivers the message with quantitative analysis, visualizations, and prototyping in cross-functional team settings.

Professional Experience

ACT Education Corp., Iowa City, IA

July 2021 – Present

Senior Research Scientist

- Enhancing CRASE, an automated essay scoring system for high-stakes writing assessments, saving $\sim$ \$1.5M annually in human scoring costs with ML/LLM modules for main scoring engine, single-score scoring logic, a disturbing-content classifier, and an off-topic detector.
- Researching methods for more efficient and effective assessment development and delivery, including LLM-based item parameter prediction models and synthetic data generation for training scoring models.
- Designing and prototyping LLM-based tools for content developers and students, including passage-based question generation and essay tutoring applications.
- Developing various internal automation solutions, including form assembly, sales region recommendation, open-ended survey response analysis, and question-solving-step generation.
- Maintaining and improving existing ML modules in production.

PEARSON, Centennial, CO

June 2020 – May 2021

Data Scientist, Intelligent Systems Research and Development (Aida Calculus product)

- Created neural-network models to represent mathematical equations and find similar items or matching templates.
- Developed an IRT-based Bayesian network learner model to estimate students' proficiency.
- Implemented RESTful APIs for the app's popular features, such as practice, diagnostic dashboard, and hint selection.

Technical Skills

Programming Skills

- **Python:** PyTorch, scikit-learn, pandas, NumPy, Matplotlib/seaborn, NetworkX, Flask.
- **R:** LME4, tm, topicmodels, bnlearn, JAGS, ggplot, Shiny.
- **Others:** Databricks, Sagemaker, Docker, AWS, GCP, Bash shell, Git, SQL.

Research Methods

- Deep neural networks, fine-tuning language models, automated prompt engineering, contrastive learning, synthetic data generation, reinforcement learning, mixed-effect analysis.
- Visualization, A/B testing, eye-tracking, mixed research methods.

Education

Ph.D., Information, University of Michigan, Ann Arbor, MI

2020

Thesis: [Understanding Language Learning via Machine Learning Methods on Behavioral and](#)

Publications

Peer-Reviewed Proceedings and Journals

- Demirtaş, M. A., **Nam, S.**, & Griffin, G. (2026). Leveraging Human-AI Collaboration for a Passage-Based Question Authoring Tool. In International Conference on Artificial Intelligence in Education (pp. 475-489).
- **Nam, S.** (2026). Prompt Optimization with Verifiable Rewards for Synthetic Essay Generation. In International Conference on Artificial Intelligence in Education (pp. 125-139).
- **Nam, S.** (2025). Domain-Adaptive Automated Essay Scoring with Topic Relevance Learning. In Proceedings of the Second Workshop on Automated Evaluation of Learning and Assessment Content co-located with 26th International Conference on Artificial Intelligence in Education (AIED 2025).
- **Nam, S.** (2025). Ordinal Classification for Transformer-Based Automated Essay Scoring Models. In International Conference on Artificial Intelligence in Education (pp. 252-259).
- Yang, K. B., **Nam, S.**, Huang, Y., & Wood, S. (2024). Rhetor: Providing LLM-Based Feedback for Students' Argumentative Essays. In European Conference on Technology Enhanced Learning (pp. 201-205).
- **Nam, S.**, Collins-Thompson, K., Jurgens, D., & Tong, X. (2024). Finding Educationally Supportive Contexts for Vocabulary Learning with Attention-Based Models. In Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024) (pp. 7286-7295).
- Yang, B., **Nam, S.**, & Huang, Y. (2023). "Why My Essay Received a 4?": A Natural Language Processing Based Argumentative Essay Structure Analysis. In International Conference on Artificial Intelligence in Education (pp. 279-290).
- **Nam, S.**, Bylinskii, Z., Tensmeyer, C., Jain, R., Wigington, C., & Sun, T. (2020). Why are you reading this? Predicting reading goal and familiarity from people's mobile interaction behaviors. *Journal of Vision*, 20(11), 951-951.
- **Nam, S.**, Samson, P. (2019). Integrating Students' Behavioral Signals and Academic Profiles in Early Warning System. In International Conference on Artificial Intelligence in Education, 345-357.
- **Nam, S.**, Frishkoff, G. A., & Collins-Thompson, K. (2018). Predicting Students' Disengaged Behaviors in an Online Meaning-Generation Task. In *IEEE Transactions on Learning Technologies* 11 (3), 362-375.
- **Nam, S.**, Collins-Thompson, K., & Frishkoff, G. A. (2017). Predicting Short- and Long-Term Student Learning with a Vocabulary Tutoring System via Semantic Features of Partial Word Knowledge. In *Educational Data Mining*, 80-87.
- Waddington, R. J., **Nam, S.**, Lonn, S., Teasley, S. D. (2016). Improving Early Warning Systems with Categorized Course Resource Usage. In *Journal of Learning Analytics* 3 (3), 263-290.
- **Nam, S.** (2016). Predicting Off-task Behaviors in an Adaptive Vocabulary Learning System. In *Educational Data Mining*, 672-674. (Doctoral Consortium)
- Oh, J., **Nam, S.**, & Lee, J. (2014). Generating highlights automatically from text-reading behaviors on mobile devices. In *CHI'14 Extended Abstracts on Human Factors in Computing*

Systems (pp. 2317-2322).

- **Nam, S.**, Lonn, S., Brown, T., Davis, C. S., & Koch, D. (2014). Customized course advising investigating engineering student success with incoming profiles and patterns of concurrent course enrollment. In Proceedings of the Fourth International Conference on LAK. ACM, 16-25.
- Waddington, R. J., & **Nam, S.** (2014). Practice exams make perfect: incorporating course resource use into an early warning system. In Proceedings of the Fourth International Conference on LAK. ACM, 188-192.

Book Chapters

- **Nam, S.**, Collins-Thompson, K., & Frishkoff, G. A. (2017). Modeling Off-task Behaviors and Learning Outcomes on a Meaning-Generation Task. In Analytics for Learning White Paper: Measurement in digital environments.
- Frishkoff, G. A., Collins-Thompson, K., **Nam, S.**, Hodges, L., & Crossley, S. (2016). Dynamic Support of Contextual Vocabulary Acquisition for Reading (DSCoVAR): An intelligent tutor for contextual word learning. In Adaptive Educational Technologies for Literacy Instruction. N.Y., NY: Routledge.

Preprints and Technical Reports

- Wood, S., Li, D., & **Nam, S.** (2026). Validity Arguments for Using CRASE[®] Confidence Scoring on ACT Writing Test Essays. Research Report. R2611.
- Wood, S., **Nam, S.**, & Li, D. (2026). CRASE5[®] for ACT Writing Technical Report. Research Report. R2544.
- **Nam, S.**, Jurgens, D., & Collins-Thompson, K. (2022). An Attention-Based Model for Predicting Contextual Informativeness and Curriculum Learning Applications. arXiv preprint arXiv:2204.09885.
- **Nam, S.**, Bylinskii, Z., Tensmeyer, C., Wigington, C., Jain, R., & Sun, T. (2020). Using Behavioral Interactions from a Mobile Device to Classify the Reader's Prior Familiarity and Goal Conditions. arXiv preprint arXiv:2004.12016.

Community Service

- **Reviewer Service:**
 - NLP: EMNLP, ACL, BEA
 - HCI: ACM CHI, ACII, UMUI
 - Educational Technology: AIED, ACM LAK, ACM L@S, BJET, JATT, EM: I.P., AIME-Con.
- Student Organizer of Academic Innovation at Michigan Analytics Talk Series *Jan 2016 – May 2019*
- Local Arrangement Chair for ACM SIGIR *2018*

Invited Talks

- Machine Learning Research in Educational Technology (October 2023) at the University of Wisconsin, Computer Science Department.
- Learning from Implicit Signals in Learning (August 2018) at Seoul National University, Department of Psychology.

- Understanding Online Behavior and Performance Signals in Educational Systems (April 2018) at the University of Oregon, Computer and Information Science Department.
- Predicting Short- and Long-Term Student Learning with a Vocabulary Tutoring System via Semantic Features of Partial Word Knowledge (June 2017) at Seoul National University, Department of Communication.
- Learning Analytics with Mass Data: Analyzing Higher Education's Academic Performance and Decision Making with Data (June 2014) at Seoul National University, Department of Psychology, and Department of Communication.

Public Presentations

- **Nam, S.** (2026). Synthetic Data Generation for Automated Essay Scoring Models. Presented at National Council on Measurement in Education Annual Meeting, Los Angeles, CA.
- **Nam, S.** (2025). Improving Zero-Shot Classification of Alerting Content Essays with Automatically Generated Descriptions. Presented at National Council on Measurement in Education Annual Meeting, Denver, CO.
- **Nam, S.** (2025). Hybrid Approach for Understanding Educational Surveys Using Generative Model and Text Encoders. Presented at National Council on Measurement in Education Annual Meeting, Denver, CO.
- **Nam, S., & Samson, P.** (2018). Mining Students' In- and Out-of-class Behaviors to Create Earlier Warning System. Presented at Michigan Institute for Data Science Annual Data Science Symposium, Ann Arbor, MI.
- **Nam, S., & Collins-Thompson, K.** (2018). Toward Understanding Contextual Information in Text. Presented at University of Michigan Data Science Learning Analytics Challenge Symposium, Ann Arbor, MI.
- **Nam, S., Collins-Thompson, K., & Frishkoff, G. A.** (2017). Predicting Short- and Long-Term Student Learning with a Vocabulary Tutoring System via Semantic Features of Definition Responses. Presented at University of Michigan Data Science Research Forum, Ann Arbor, MI.
- **Nam, S., & Collins-Thompson, K.** (2017). Understanding Questions with Object and Action Terms: Bi-modal topic modeling of student questions from Piazza. Presented at Michigan Institute for Data Science Annual Symposium, Ann Arbor, MI.
- **Nam, S., Collins-Thompson, K., Frishkoff, G. A., Bhide, A., Muth, K. B., & Perfetti, C.** (2016). Modeling Real-time Performance on a Meaning-Generation Task. Presented in American Educational Research Association Annual Meeting, Washington, D.C.; Michigan Institute for Computational Discovery & Engineering Symposium, Ann Arbor, MI.
- **Oh, J., Nam, S., & Lee, J.** (2013). Enhancing Effectiveness of Skim Reading with Highlighting Text. Presented in HCI Korea 2013, Seoul, South Korea.

Research Internship and Assistant Experiences

ADOBE, San Jose, CA

June – Nov 2019

Machine Learning Intern, Document Intelligence Lab (Mentor: Dr. Tong Sun)

- Designed an experiment to predict mobile device readers' prior knowledge levels and goal conditions from scrolling behaviors. The experiment included a crowdsourcing task with a web application, analyzing eye-tracking signals, user interviews, and developing a mixed-effect classifier.

ECHO 360, Ann Arbor, MI

May – Aug 2018

Graduate Research Intern (Mentor: Dr. Perry Samson)

- Developed early warning prediction model from behavioral logs and academic profiles and suggested customized learning strategies for different student groups.
- Supervised undergraduate interns in developing software for transcription, text search, and text clustering.

UNIVERSITY OF MICHIGAN, Ann Arbor, MI

Sep 2012 – May 2020

Graduate Research Assistant

- Developed a deep learning model to identify more instructive sentences for a vocabulary learning system and examined how sentences improved efficiency in training NLP models like word2vec or FastText.
- Created an interpretable semantic scale for text by projecting biases in semantic differential with word2vec.
- Implemented a prediction model for students' disengaged behaviors and intervention moments.
- Expanded a topic model with grammar dependency on online forum texts for richer representation.
- Analyzed students' successful learning behaviors, like exploring course-taking patterns from >10 years of enrollment data or behavioral logs from multiple STEM courses' learning management systems.